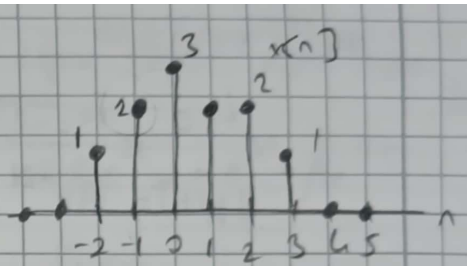
2.31)

$y[n] + 2y[n-1] = x[n] + 2x[n-2]$

$y[n] = x[n] + 2x[n-2] - 2y[n-1]$



* $y[-2] = x[-2] + 2x[-4] - 2y[-3]$

$= 1 + 0 - 0$ → başlangıçta = 0

$y[-2] = 1$

* $y[-1] = x[-1] + 2x[-3] - 2y[-2]$

$= 2 + 0 - 2$

$y[-1] = 0$

* $y[0] = x[0] + 2x[-2] - 2y[-1]$

$= 3 + 2 - 0$

$= 5$

* $y[1] = x[1] + 2x[-1] - 2y[0]$

$= 2 + 4 - 10$

$y[1] = -4$

* $y[2] = x[2] + 2x[0] - 2y[1]$

$= 2 + 6 + 8$

$y[2] = 16$

* $y[3] = x[-3] + 2x[1] - 2y[2]$

$= 0 + 4 - 4$

$= 0$

* $y[4] = 58$

* $y[5] = -114(-2)^{n-5}$

$$y[n] = \begin{cases} 0, & n \leq -3 \\ 1, & n = -2 \\ 0, & n = -1 \\ 5, & n = 0 \\ -4, & n = 1 \\ 16, & n = 2 \\ -22, & n = 3 \\ 58, & n = 4 \\ -114(-2)^{n-5} & n = 5 \end{cases}$$

2.33 $\frac{dy(t)}{dt} + 2y(t) = x(t)$

i) output $y_1(t)$, input is $x_1(t) = e^{3t}u(t)$.

$y_1(t) = y_p(t) + y_h(t)$

Particular → $y_p(t)$
homogeneous → $y_h(t)$

Homogeneous: assume $\frac{d}{dt} = s$ and $x(t) = 0$ (input signal)

$y_h(t) = Ae^{st}$

$Ae^{st}(s) + 2Ae^{st} = 0$

$Ae^{st}(s+2) = 0$

↳ 0 $\boxed{s = -2}$

$y_h(t) = Ae^{-2t}$

Particular: $y_p(t) = Ke^{3t}u(t)$

$\frac{d}{dt}(Ke^{3t}u(t)) + 2Ke^{3t}u(t) = e^{3t}u(t)$

$3Ke^{3t} + 2Ke^{3t} = e^{3t}$

$5Ke^{3t} = e^{3t}$

$\boxed{K = 1/5}$

$y_p(t) = \frac{1}{5}e^{3t}u(t) \rightarrow \frac{1}{5}e^{3t}$ for $t > 0$

$y_1(t) = y_p(t) + y_h(t)$

$= \frac{1}{5}e^{3t} + Ae^{-2t}$ for $t > 0$

system is initially rest

$y(0) = 0$ $t < 0$ $y(t) = 0$

$y(0) \Rightarrow \frac{1}{5} + A = 0$

$A = -\frac{1}{5}$

result is $\Rightarrow \left[\frac{1}{5}e^{3t} - \frac{1}{5}e^{-2t}\right]u(t)$

ii) output $y_2(t)$, input is $x_2(t) = e^{2t}u(t)$

$y_2(t) = y_p(t) + y_h(t)$

Homogeneous: assume $\frac{d}{dt} = s$ and $x(t) = 0$

$y_h(t) = Ae^{st}$

$Ae^{st}(s) + 2Ae^{st} = 0$

$Ae^{st}(s+2) = 0$

$s = -2$

$\boxed{y_h(t) = Ae^{-2t}}$

Particular: $y_p(t) = Ke^{2t}u(t)$

$\frac{d}{dt}\left[Ke^{2t}u(t)\right] + 2Ke^{2t}u(t) = e^{2t}u(t)$

$2Ke^{2t} + 2Ke^{2t} = e^{2t}$

$4K = 1$

$K = 1/4$

$y_p(t) = \frac{1}{4}e^{2t}u(t)$

↳ $\frac{1}{4}e^{2t}$ for $t > 0$

$y_2(t) = y_p(t) + y_h(t)$

$= \frac{1}{4}e^{2t} + Ae^{-2t}$ for $t > 0$

$t = 0$ $y(t) = 0$ → $\frac{1}{4} + A = 0$

↳ $-\frac{1}{4}$

result is $\Rightarrow \left[\frac{1}{4}e^{2t} - \frac{1}{4}e^{-2t}\right]u(t)$

000 LTI output $y_3(t)$ input $x_3(t) = \alpha e^{3t}u(t) + \beta e^{2t}u(t)$ where α and β real number

Show that $y_3(t) = a y_1(t) + \beta y_2(t)$

$y_3(t) = y_p(t) + y_h(t)$

Homogenous: $x(t) = 0$

$y_h(t) = Ae^{st}$ $\quad \frac{d}{dt} = s$

$\frac{d}{dt}(Ae^{st}) + 2Ae^{st} = 0$

$As.e^{st} + 2Ae^{st} = 0$

$(s+2) = 0$

$s = -2$

$y_h(t) = Ae^{-2t}$

Particular: $y_p(t) = K_1\alpha e^{3t}u(t) + K_2\beta e^{2t}u(t)$

$\frac{d}{dt}\left[K_1\alpha e^{3t}u(t) + K_1\beta e^{2t}u(t)\right] + 2K_2\alpha e^{3t}u(t) + 2K_2\beta e^{2t}u(t) = \alpha e^{3t}u(t) + \beta e^{2t}u(t)$

$3K_1\alpha e^{3t} + 2K_1\beta e^{2t} + 2K_2\alpha e^{3t} + 2K_2\beta e^{2t} = \alpha e^{3t} + \beta e^{2t}$ for $t>0$

$5K_1(\alpha e^{3t}) + 4K_2\beta e^{2t} = \alpha e^{3t} + \beta e^{2t}$

$K_1 = \frac{1}{5}$ $\quad K_2 = \frac{1}{4}$

$y_p(t) = \frac{1}{5}\alpha e^{3t}u(t) + \frac{1}{4}\beta e^{2t}u(t)$

$y_3(t) = Ae^{-2t} + \frac{1}{5}\alpha e^{3t} + \frac{1}{4}\beta e^{2t}$

$y_3(0) = 0$

$y_3(0) \Rightarrow A + \frac{\alpha}{5} + \frac{\beta}{4} = 0$ $\quad A = \frac{-\alpha}{5} - \frac{\beta}{4}$

result is $\left(\left(\frac{-\alpha}{5} - \frac{\beta}{4}\right)e^{-2t} + \frac{1}{5}\alpha e^{3t} + \frac{1}{4}\beta e^{2t}\right) =$

$y_1(t) = \left(-\frac{1}{5}e^{-2t} + \frac{1}{5}e^{3t}\right)u(t)$

$y_2(t) = \frac{1}{4}\left(e^{2t} - e^{-2t}\right)u(t)$

so we can't write

$y_3(t) = \alpha y_1(t) + \beta y_2(t)$

iv) input $x_1(t)$ output $y_1(t)$
input $x_2(t)$ output $y_2(t)$

$x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$\frac{dy(t)}{dt} + 2y(t) = x(t)$

$x_1(t) = \frac{dy_1(t)}{dt} + 2y_1(t)$

$x_2(t) = \frac{dy_2(t)}{dt} + 2y_2(t)$

$\alpha x_1(t) \longrightarrow \alpha y_1(t)$

$\beta x_2(t) \longrightarrow \beta y_2(t)$

$x_3(t) = \alpha y_1(t) + \alpha y_2(t)$

we can say system is linear.

b

$\frac{dy(t)}{dt} + 2y(t) = x(t)$

i)

$y_1(t) = y_h(t) + y_p(t)$

input $\Rightarrow x_1(t) = Ke^{2t}u(t)$

Homogeneous input signal $x_1(t) = 0$ $\left[\frac{d}{dt} = s\right]$ $y_h(t) = Ae^{st}$

$\Rightarrow \frac{dy_h(t)}{dt} + 2y_h(t) = 0$

$\Rightarrow s\,y_h(t) + 2y_h(t) = 0$

$2 + s = 0$

$s = -2$

$y_h(t) = Ae^{-2t}$ for $t > 0$

Particular input $x_1(t) = Ke^{2t}u(t)$ for $t > 0$

$y_p(t) = k_1(ke^{2t})$

$\frac{dy_p(t)}{dt} + 2y_p(t) = ke^{2t}$

$2k_1 ke^{2t}(2) + 2k_1 ke^{2t} = ke^{2t}$

$2k_1 + 2k_1 = 1$

$k_1 = \frac{1}{4}$ $\quad y_p(t) = \frac{k}{4}e^{2t}$

$y_1(t) = Ae^{-2t} + \frac{k}{4}e^{2t}$ $t > 0$ $\quad y_1(0) = 0$

$0 = A + \frac{k}{4}$

$-\frac{k}{4} = A$

result is $\Rightarrow$ $y_1(t) = \left[-\frac{k}{4}e^{-2t} + \frac{k}{4}e^{2t}\right]u(t)$

ii) input $\Rightarrow x_2(t) = k e^{2(t-T)} u(t-T)$

Homogeneous $\frac{d}{dt} = S \quad y_h(t) = Ae^{st}$

$\Rightarrow \frac{d\,y_h(t)}{dt} + 2y_h(t) = 0$

$\Rightarrow S\,y_h(t) + 2y_h(t) = 0$

$2 + S = 0$

$S = -2$

$y_h(t) = Ae^{-2t}$ for $t > 0$

Particular input $x_2(t) = k e^{2(t-T)}$ for $t > T$

$y_p(t) = k_2 \cdot k e^{2(t-T)}$

$\frac{d}{dt}\left[k_2 \cdot k e^{2(t-T)}\right] + 2 \cdot k_2 \cdot k e^{2(t-T)} = k e^{2(t-T)}$

$\Rightarrow 2k k_2 e^{2(t-T)} + 2k_2 \cdot k e^{2(t-T)} = k e^{2(t-T)}$

$k_2 = \frac{1}{4} \qquad y_p(t) = \frac{k}{4} e^{2(t-T)}$

$y_2(t) = Ae^{-2t} + \frac{k}{4} e^{2(t-T)}$ for $t > T$

initially rest $\quad y_2(T) = 0$

$0 = Ae^{-2T} + \frac{k}{4}$

$A = -\frac{k}{4} e^{2T}$

$y_2(t) = \frac{k}{4}\left[e^{2(t-T)} - e^{-2(t-T)}\right] u(t-T)$

$y_1(t) = \frac{k}{4}\left[e^{2t} - e^{-2t}\right] u(t)$

$\Rightarrow y_1(t-T) = \frac{k}{4}\left[e^{2(t-T)} - e^{-2(t-T)}\right] u(t-T)$

$\Rightarrow y_2(t) = y_1(t-T)$

iii)

$x_1(t) = 0$ for $t < t_0$

$x_1(t) \Rightarrow y_1(t)$

$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1(t)$

$y_1(t) = 0$ for $t < t_0$

$y_2(t) = y_1(t-T)$

$\Rightarrow y_2(t) = 0$ for $t < t_0 + T$

$x_2(t)$

$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2(t)$

$x_2(t) = x_1(t-T)$

$\frac{dy_1(t-T)}{dt} + 2y_1(t-T) = x_1(t-T)$

system is time-invariant